

LOUCA MALERBA

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EDUCATION

CentraleSupélec, Université Paris-Saclay

2023 – 2026

Top 3 Engineering School in France, part of 13th-ranked university worldwide (Shanghai Ranking)

Master's Degree (M2) in General Engineering - Data Science Specialization — GPA: 3.96/4.33

Coursework: Machine Learning, Deep Learning, Optimization, Statistical Modeling, HPC

CPGE Jules Ferry

2021 – 2023

Preparatory Classes - PTSI/PT — Ranked 5th nationwide - Top 5% of students*

Intensive foundation program in mathematics, physics, engineering — National competitive exam preparation

EXPERIENCE

GE HealthCare, Buc, France

Oct 2023 – Sept 2026 (Expected)

Apprentice Research Engineer

- Developing end-to-end deep learning pipeline for chronic pulmonary embolism (CTEPH) detection and segmentation in 3D CT scans as part of *Destination 2024* partnership between GE HealthCare and AP-HP, from literature review to clinical validation on hospital data warehouses.
- Benchmarking and adapting SOTA frameworks (nnUNet, nnDetection) with custom preprocessing pipelines. Investigating model robustness on noisy clinical data.
- **Targeting a publication at MICCAI** based on novel classification methodologies developed during this research.

PROJECTS

Dual-sPLS: a family of Dual Sparse Partial Least Squares regressions for feature selection and prediction with tunable sparsity ([link](#))

Independent Research Collaboration with Laurent Duval, Adjunct Professor at CentraleSupélec

- Acknowledged contribution in forthcoming paper (CalValXy)
- Contributor of work in progress (Dual-sPLS)

Active Learning for Plankton Segmentation — Side Project ([link](#))

- Developed a human-in-the-loop pipeline to segment images under significant domain shift
- Implemented an uncertainty sampling strategy with stochastic batches

More projects are in progress or available on my portfolio ([link](#))

TECHNICAL SKILLS

Languages & Software Engineering: Python, C++, Git

Deep Learning & ML: PyTorch, Scikit-learn, Computer Vision, 3D Deep Learning

Infrastructure & Distributed Computing: Linux, SLURM

Computer Vision for Medical AI: nnUNet, nnDetection, SimpleITK, 3D Slicer

LANGUAGES

French: Native — **English:** Fluent (C1 – TOEFL 111/120) — **Italian:** Intermediate